Project Report: Heart Disease Prediction

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1. Problem Statement

Cardiovascular diseases are a significant global health challenge. Organizations require structured, data-driven insights for strategic clinical decisions. This project aims to solve the real-world problem of predicting heart disease likelihood using a complete data science workflow, identifying key physiological risk factors to enable early intervention and improved patient outcomes.

2. Project Overview

This report performs an end-to-end analysis of the `heart.csv` dataset. We move from initial data cleaning and exploratory data analysis (EDA) to identifying correlations and preparing the data for analytical modeling.

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GitHub Repository: https://github.com/mostlystoic/End-to-End-Data-Science-Case-Study-And-Report
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3. Tools & Technologies

- **Programming Language:** Python
- **Data Libraries:** Pandas, NumPy
- **Visualization:** Matplotlib, Seaborn
- **Reporting:** WeasyPrint (HTML to PDF conversion)

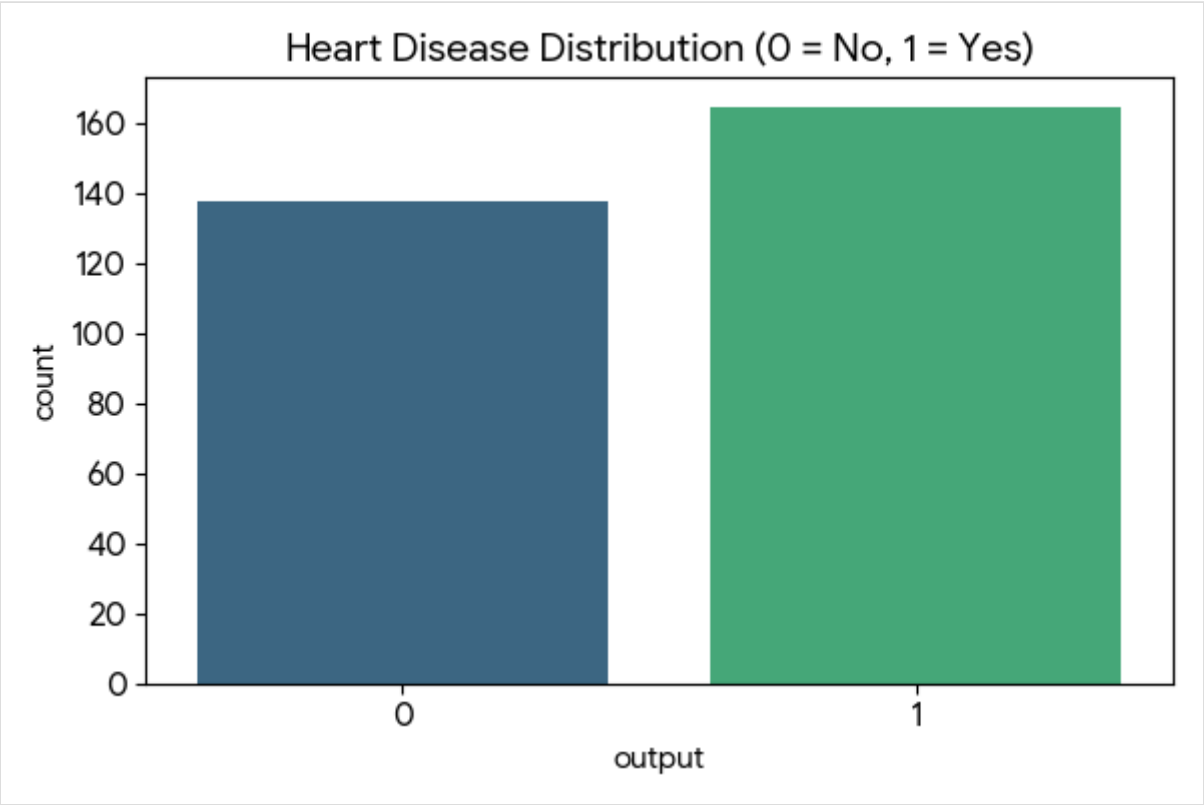
4. Data Exploration & Cleaning

The dataset consists of 303 rows and 14 columns. Data cleaning involved checking for null values (none found) and verifying data types. Below is a statistical summary of the clinical features:

	age	sex	cp	trtbps	chol	fbs	restecg	thalachh
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000
mean	54.366337	0.683168	0.966997	131.623762	246.264026	0.148515	0.528053	149.646865
std	9.082101	0.466011	1.032052	17.538143	51.830751	0.356198	0.525860	22.905161
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000	71.000000
25%	47.500000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000	133.500000
50%	55.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000	153.000000
75%	61.000000	1.000000	2.000000	140.000000	274.500000	0.000000	1.000000	166.000000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000	202.000000

Visual Exploration: Target Distribution

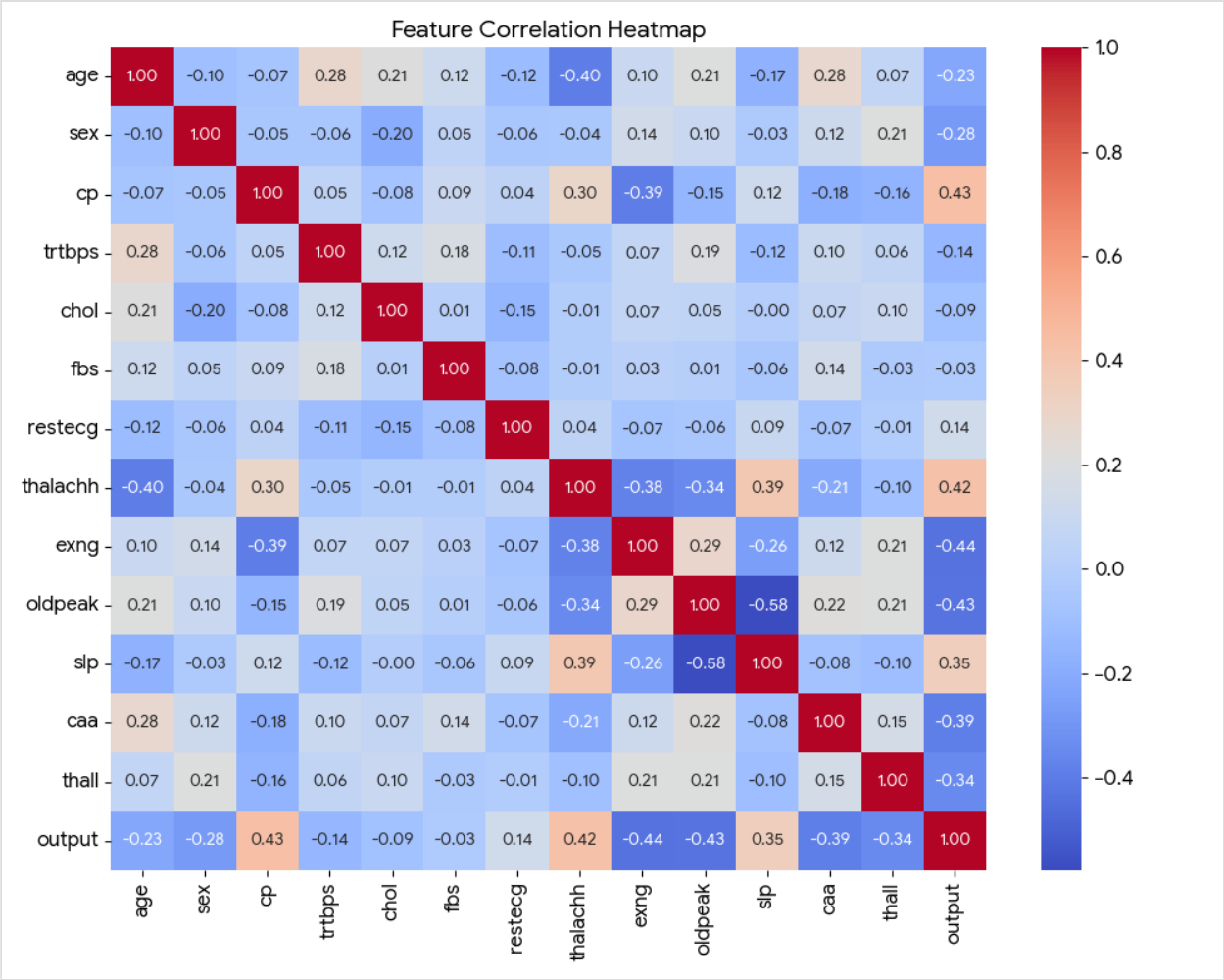
The dataset is relatively balanced, with a slightly higher prevalence of heart disease cases (1) compared to healthy cases (0).


Heart Disease Distribution (0 = No, 1 = Yes)
count
160
140
120
100
80
60
40
20
0
0
1
output

5. Analytical Techniques

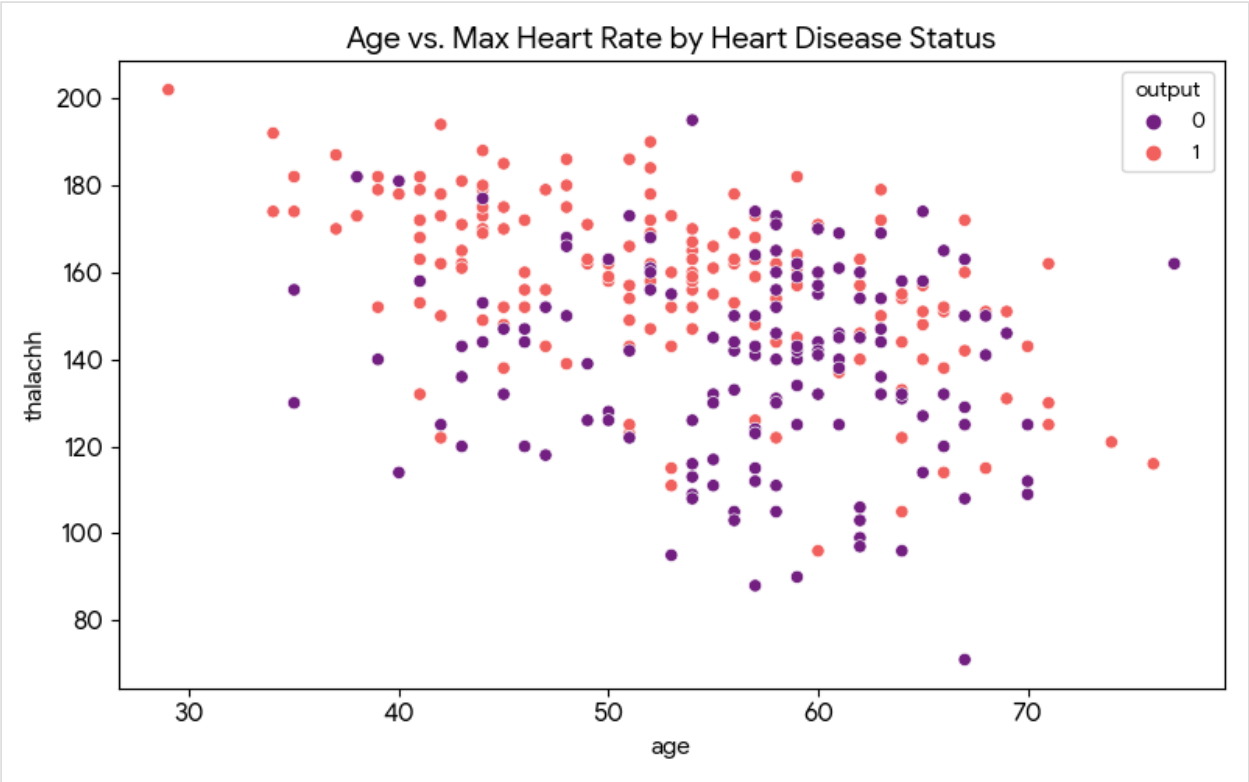
Feature Correlation Analysis

We applied correlation heatmaps to understand the relationship between variables. High positive correlation is observed between chest pain (cp) and the output, while exercise-induced angina (exng) shows a strong negative correlation.



Bivariate Analysis: Age and Heart Rate

Exploring the relationship between age and maximum heart rate achieved (`thalachh`) shows that patients with heart disease generally exhibit higher maximum heart rates across various age groups.



6. Conclusion

This end-to-end analysis demonstrates that clinical features such as chest pain type and maximum heart rate are vital predictors of heart disease. The structured insights generated here provide a foundation for building automated diagnostic support tools that can enhance clinical decision-making.